

Intrinsic Information Limit in Nuclear Optical Potential Extraction

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How many independent parameter combinations can nucleon-nucleus scattering data actually constrain in the optical model potential? Using the effective dimensionality D_{eff} (the participation ratio of Fisher information eigenvalues), I show that, within the Koning-Delaroche (KD02) framework, the continuous “Igo ambiguity” reflects a structural information bottleneck rather than simply insufficient data. Single-energy elastic scattering constrains only $D_{\text{eff}} \approx 1.7 \pm 0.4$ parameter combinations (averaged over 168 configurations spanning $A = 12\text{--}208$ and $E = 10\text{--}200$ MeV). Decomposing how each data class redistributes the Fisher spectrum reveals a clear hierarchy for opening independent parameter directions: systematic multi-energy fitting within the KD02 energy dependence gives the largest increase ($D_{\text{eff}} \rightarrow \sim 2.3\text{--}2.5$ in the 13-parameter local space; $D_{\text{eff}} \approx 2.9$ out of 48 universal KD02 coefficients), the analyzing power A_y gives the next-largest increase for most neutron configurations at representative precision ratios, while σ_R alone opens essentially no new direction. Verified across 12 nuclei, 7 energies, and both projectile types, this bottleneck is stable across the configurations sampled: uniform error reduction rescales the local Fisher spectrum without changing its anisotropy, whereas qualitatively different measurements can partially broaden it. For downstream reaction calculations, this identifies which optical-potential directions must be pinned or propagated before discrepancies are assigned to reaction or structure physics. In logarithmic KD02 coordinates, the spectrum has the standard local signature of a “sloppy model.” The analysis provides a systematic effective-dimensionality map of the Igo ambiguity and establishes the priority ordering for opening independent directions beyond elastic scattering: multi-energy systematics $\gg A_y \gg \sigma_R$.

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Introduction.—Direct reactions on weakly bound and rare isotopes infer structure and response information through optical potentials that are not observables themselves. When a transfer, breakup, or charge-exchange calculation disagrees with data, the first diagnostic question is whether the discrepancy originates in the optical interaction, the reaction mechanism, or the structure input. This question reaches back to the inception of the optical model potential (OMP) over seventy years ago [1–3]. Since then, the OMP has become the effective interaction behind much of nuclear reaction theory [4–7], but a basic inverse-problem obstruction persists: the “inverse problem ill-posedness,” historically known as the Igo ambiguity [8]. Distinct parameter sets, representing different nuclear potentials, can yield indistinguishable scattering cross sections [9, 10], creating a continuous degeneracy that defies unique determination.

Two types of ambiguity exist [5]: *discrete* ambiguity from different wavefunction node numbers (resolvable by high-energy rainbow scattering), and *continuous* ambiguity where parameters trade off while maintaining the potential near the nuclear surface [10]. For a Woods-Saxon potential, this surface sensitivity is captured by the volume integral $J_V \propto V r_v^3$ [5]. The qualitative origin of this continuous ambiguity is well understood: elastic scattering is primarily sensitive to the potential near the nuclear surface, making it insensitive to the detailed radial shape [5]. However, despite this qualitative under-

standing, three key questions lack quantitative answers: (1) *How many* independent parameter combinations can actually be constrained? (2) Can improved experimental precision eventually resolve the degeneracy? (3) Is the ambiguity universal across the nuclear chart?

Recent Bayesian uncertainty quantification [11, 12] has documented large parameter uncertainties in nuclear reaction models. The motivation goes beyond attaching error bars to optical-model fits: in transfer, breakup, charge-exchange, and other reactions involving weakly bound or rare isotopes, a disagreement with data cannot be interpreted until the error budget separates optical-potential input from reaction-mechanism approximations, structure overlaps, and missing channels [13, 14]. This need is explicit in the rare-isotope-beam literature, where quantifying and reducing reaction-model uncertainties, especially those associated with nuclear optical potentials, has been identified as a targeted priority [15]. In a landmark comparison, King *et al.* [16] showed that Bayesian MCMC posterior distributions for optical potential parameters are strongly non-Gaussian, with only the depth V and radius r_v strongly correlated, in striking contrast to the frequentist χ^2 approach which produced spurious correlations among many parameter pairs. Subsequent studies confirmed these findings [13, 14, 17], while principal component analysis identified the dominant variance directions [18]. Yet these empirical observations require $\sim 100,000$ MCMC samples per system

for converged correlation studies and lack a theoretical framework that would answer the questions above.

In this Letter, I quantify these questions using Fisher information geometry [19], focusing on the continuum regime ($E = 10\text{--}200$ MeV), above the resonance region where smooth optical potentials such as KD02 are designed to apply. In logarithmic KD02 coordinates and for the stated error models, the eigenvalue spectra have the standard signature of “sloppy models” [20–24]: only a few “stiff” (large eigenvalue) parameter combinations are well constrained, while the remaining directions are “sloppy” (data-insensitive). This is a local spectral classification, not a requirement that published best-fit parameters drift by orders of magnitude; physical constraints, shared parameter relations, and global systematics can pin directions that one observable class leaves flat. By systematically scanning 168 configurations across the nuclear chart and progressively adding observables, I establish the stability of this bottleneck and identify which data classes open independent parameter directions beyond elastic scattering.

Method.—Consider nucleon-nucleus elastic scattering with the Koning-Delaroche (KD02) global optical potential [25–27], which has 9 parameters for the central terms: real volume (V, r_v, a_v), imaginary volume (W, r_w, a_w), and imaginary surface (W_d, r_d, a_d). Including spin-orbit coupling with Thomas form adds 4 parameters ($V_{so}, W_{so}, r_{so}, a_{so}$), yielding a 13-parameter model for spin-1/2 scattering. (Although KD02 constrains $r_w = r_v$ and $a_w = a_v$, I retain them as independent in the Fisher analysis to test whether data could in principle distinguish them; see End Matter.)

The *relative sensitivity* of the differential cross section $\sigma(\theta) \equiv d\sigma/d\Omega(\theta)$ to each parameter p_i is

$$S_i(\theta) = \frac{\partial \log \sigma(\theta)}{\partial \log p_i} = \frac{p_i}{\sigma} \frac{\partial \sigma}{\partial p_i}. \quad (1)$$

Using logarithmic derivatives makes parameters with different physical units directly comparable. Working in logarithmic parameter space throughout, the Fisher information matrix [19] (FIM) for a data vector $\{y_k\}$ with uncertainties $\{\delta_k\}$ is

$$F_{ij} = \sum_k \delta_k^{-2} \frac{\partial y_k}{\partial \log p_i} \frac{\partial y_k}{\partial \log p_j}, \quad (2)$$

where $\partial y_k / \partial \log p_i = p_i \partial y_k / \partial p_i$. For elastic cross sections with uniform relative uncertainties $\delta_k = \epsilon \sigma(\theta_k)$, this reduces to $F_{ij} = \epsilon^{-2} \sum_\theta S_i(\theta) S_j(\theta)$. The overall factor ϵ^{-2} cancels in D_{eff} (see below), so the information structure depends only on the sensitivity matrix $S_i(\theta)$, evaluated at the KD02 reference parameter values for each (A, Z, E) configuration, using 35 scattering angles equally spaced from 5° to 175° (convergence verified; see End Matter). For A_y , absolute uncertainties are used ($\delta_{A_y} = 0.03$); for σ_R and σ_T , uniform relative uncertainties are assumed (σ_T enters for neutron projectiles

only, since the Coulomb divergence of $f(0)$ leaves it undefined for protons). The specific value of ϵ need not be specified: it cancels in D_{eff} , so only the *relative weighting* between observable types (set by ϵ/δ_{A_y}) affects the result. The combined-observable results below use sensitivity normalization ($\epsilon = 1$, $\delta_{A_y} = 0.03$); the quantitative A_y gain depends on ϵ/δ_{A_y} (see End Matter), but the qualitative hierarchy is robust. The value of D_{eff} depends on the *structure* of the error model (relative vs. absolute, angular coverage), not on the overall error magnitude, because it is scale-invariant. Each observable type enters additively ($F = F_{\text{elastic}} + F_{A_y} + \dots$), enabling the step-by-step decomposition below.

Diagonalizing the FIM yields eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{N_p}$ and corresponding eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{N_p}$, where N_p is the number of parameters. Each λ_i quantifies how much information the data carry along the direction $\mathbf{e}_i$ in parameter space: large λ_i means tightly constrained, small λ_i means poorly constrained. Each eigenvector $\mathbf{e}_i = (e_{i,1}, \dots, e_{i,N_p})$ is a unit vector in the N_p -dimensional space of (log-)parameters; its squared component $e_{i,k}^2$ gives the fractional contribution of parameter p_k to that mode. The effective dimensionality, defined as the participation ratio [21, 28] of the FIM eigenvalues,

$$D_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}, \quad (3)$$

ranges from 1 (one direction dominates) to the total number of parameters (all directions equally constrained); $D_{\text{eff}} \approx 2$ does not mean that only 2 directions carry *any* information, but rather that information is overwhelmingly concentrated in $\sim D_{\text{eff}}$ stiff modes. Unlike the condition number $\kappa = \lambda_{\text{max}}/\lambda_{\text{min}}$, which only measures the extremes, D_{eff} captures the full eigenvalue distribution and counts how many directions carry significant information. Crucially, D_{eff} is *scale-invariant*: reducing all experimental errors by the same factor scales every λ_i uniformly, leaving D_{eff} unchanged. Better precision shrinks the error ellipsoid but cannot change its shape (the ratio of uncertainties along stiff and sloppy directions is fixed). Only qualitatively different measurements (e.g., adding A_y to $d\sigma/d\Omega$) can alter D_{eff} by changing the relative eigenvalue spectrum (see End Matter for a concrete illustration). The use of logarithmic derivatives further ensures invariance under parameter rescaling $p_i \rightarrow c_i p_i$, so that V (in MeV) and r_v (in fm) enter on an equal footing. Numerical details are given in the End Matter.

Results.—I systematically analyze 168 configurations: 12 nuclei (^{12}C to ^{208}Pb), 7 energies (10–200 MeV), and both neutron and proton projectiles. Figure 1 displays D_{eff} as a heatmap across all configurations. The uniformity (D_{eff} typically 1–2, with occasional values reaching ~ 3 for proton scattering on light nuclei at high energy) demonstrates that the information limit is stable

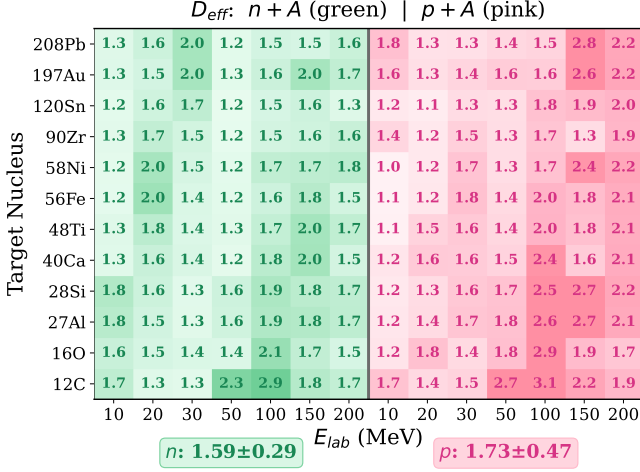


FIG. 1. Effective dimensionality D_{eff} for single-energy elastic scattering across 12 nuclei and 7 energies for neutron (green) and proton (pink) projectiles. Values typically range from 1 to 2 (occasionally reaching ~ 3), showing that only ~ 2 out of 13 KD02 parameter combinations are effectively constrained for every target and beam energy sampled.

across the configurations sampled, with only weak dependence on target mass, beam energy, or projectile type. The statistics yield $D_{\text{eff}} = 1.59 \pm 0.29$ (neutrons) and 1.73 ± 0.47 (protons), with a combined average of 1.7 ± 0.4 over all 168 configurations. The physical origin of this stability is that elastic scattering is primarily sensitive to the nuclear surface geometry, the potential radius scale, for every nucleus and beam energy sampled, as confirmed by the eigenvector analysis below.

Figure 2 confirms this stability: D_{eff} stays within 1.1–2.4 from ^{12}C to ^{208}Pb at 50 MeV [panel (a)], bounded between 1 and 3 across all energies [panel (b)], and condition numbers $\kappa = \lambda_{\text{max}}/\lambda_{\text{min}}$ of 10^6 to above 10^{10} [panel (c)], rendering the ~ 11 sloppy dimensions effectively unobservable (see End Matter).

Anatomy of the information limit.—Figure 3 shows the eigenvector composition for $n+^{40}\text{Ca}$ at 50 MeV. The dominant mode $\mathbf{e}_1$ (90% of information) is driven by r_v (62%), r_d (14%), and V (11%), corresponding to the volume integral $J_V \propto V r_v^3$; the signed eigenvector components show r_v and V entering with the same sign. The radius r_v dominates because $\partial \log J_V / \partial \log r_v = 3$ versus $\partial \log J_V / \partial \log V = 1$. The Igo ambiguity [8] corresponds to the perpendicular direction in the V - r_v plane: increasing V while decreasing r_v (or vice versa) keeps J_V approximately constant, leaving the cross section nearly unchanged. Along this direction the Fisher information is roughly a factor of 50 smaller than along $\mathbf{e}_1$. The second mode $\mathbf{e}_2$ (7%) mixes r_v (29%), a_v (27%), and r_d (24%), encoding the radial geometry. Together, these two modes capture 97% of all information; the remaining 11 directions, including diffuseness, imaginary potential, and spin-orbit, are effectively invisible to elastic angular

distributions.

The normalized sensitivity directions $\hat{S}_V(\theta)$ and $\hat{S}_{r_v}(\theta)$ overlap almost perfectly at all angles (see End Matter, Fig. 6), directly manifesting the Igo proportionality: fractional changes in V and r_v produce nearly indistinguishable effects on $d\sigma/d\Omega$. But can additional observables lift this degeneracy?

Step-by-step constraint analysis.—I track how independent parameter directions open as information sources are progressively added (Fig. 4).

Step 1: Elastic $d\sigma/d\Omega$ alone.—For $n+^{40}\text{Ca}$ at 50 MeV with 35 angular data points, $D_{\text{eff}} = 1.22$ (an illustrative case; the 168-configuration average is 1.7 ± 0.4). Only the “potential radius” mode is constrained; the eigenvalue spectrum drops by nearly eight orders of magnitude.

Step 2: Adding σ_R .—One additional datum, overwhelmed by 35 angular data points: D_{eff} increases by less than 0.001.

Step 3: Adding A_y .—The analyzing power probes spin-orbit interference through the spin-flip amplitude $g(\theta)$. Adding A_y raises D_{eff} from 1.22 to 1.53 for $n+^{40}\text{Ca}$, and from 1.15 to 1.72 for $n+^{120}\text{Sn}$ (with the observable weighting defined in Method; the gain at experimental precision ratios is smaller, see End Matter). Within the present absolutely-normalized setup, integrated cross sections (σ_R, σ_T) provide no measurable additional gain.

Step 4: Systematic multi-energy fitting.—The combined Fisher matrix is constructed by projecting each single-energy $F(E)$ onto the shared KD02 global coefficients via the Jacobian $J(E) = \partial \boldsymbol{\theta}_{\text{local}}(E) / \partial \boldsymbol{\theta}_{\text{global}}$ and summing $F_{\text{global}} = \sum_E J^T F(E) J$, then projecting back to the local space at a reference energy, $F_{\text{ref}} = J(E_{\text{ref}}) F_{\text{global}} J(E_{\text{ref}})^T$ (a stated convention; see End Matter). Combining 7 energies (10–200 MeV) with elastic data alone raises D_{eff} to 2.47 ($n+^{40}\text{Ca}$) and 2.19 ($n+^{120}\text{Sn}$), the largest single improvement, exploiting the energy lever arm: real potential dominates at low energy, absorption at high energy. Including all observables gives $D_{\text{eff}} = 2.16$ (Ca) and 2.27 (Sn); the combined value can fall below or above the elastic-only one, depending on whether A_y concentrates or broadens the spectrum for that system (see End Matter). This gain relies jointly on the additional data and the KD02 energy dependence that ties parameters across energies.

Step 5: KD02 global systematics.—The KD02 global model parametrizes each local potential parameter as a function of mass A , charge Z , and energy E through 48 universal coefficients (9 shared geometry, 16 shared depth, 10 neutron-specific, and 13 proton-specific; see End Matter, Table I). Projecting the 168 single-system Fisher matrices onto this universal space via the Jacobian $J_i = \partial \log \boldsymbol{\theta}_{\text{local}} / \partial \log \boldsymbol{\theta}_{\text{universal}}$ and summing (see End Matter, Fig. 5) yields $D_{\text{eff}} = 2.9$ (out of 48), with the dominant eigenvector (44% of information) mixing r_{v0} (70%) and $r_{so,0}$ (24%). With neutron data alone, D_{eff} peaks at ~ 3.1 after the first few systems and settles

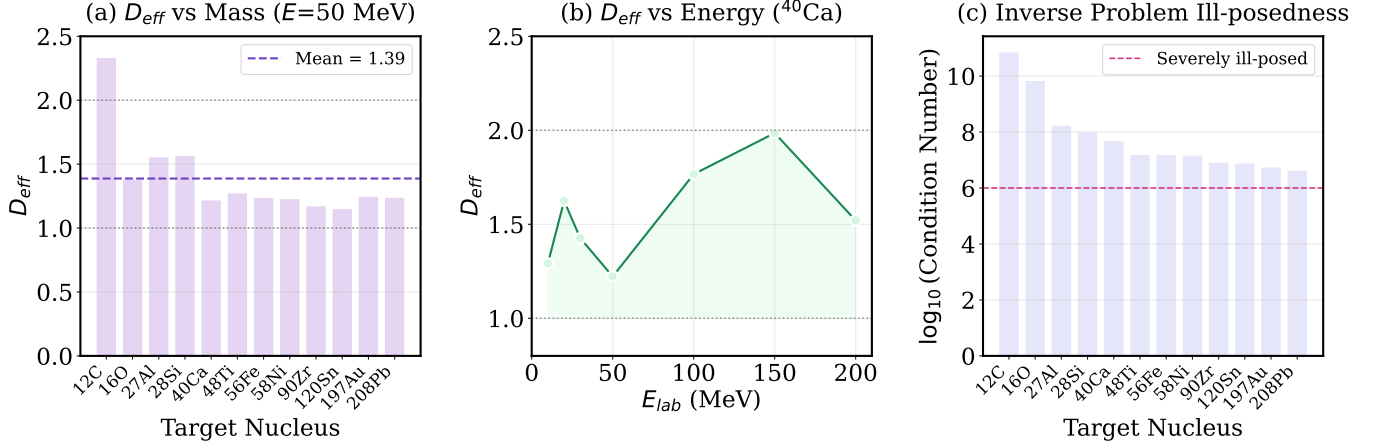


FIG. 2. Systematic analysis for $n + A$ scattering. (a) D_{eff} versus mass at $E = 50$ MeV (neutrons only; mean lower than the all-energy, all-projectile average of 1.7 ± 0.4): D_{eff} stays within 1.1–2.4 across $A = 12$ –208, largest for the lightest target. (b) D_{eff} versus energy for $n + ^{40}\text{Ca}$ stays bounded. (c) Condition number ranges from 10^6 to above 10^{10} , indicating extreme hierarchy between constrained and unconstrained directions.

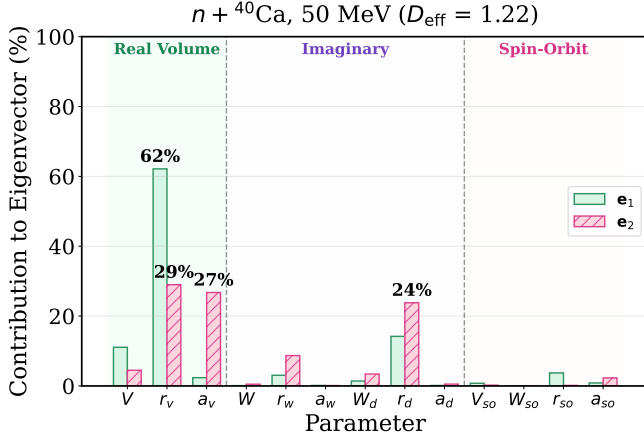


FIG. 3. Eigenvector composition for elastic $n + ^{40}\text{Ca}$ at 50 MeV (13 parameters). The “potential radius” mode $\mathbf{e}_1$ (green) is dominated by r_v (62%) with r_d (14%) and V (11%); the “radial geometry” mode $\mathbf{e}_2$ (rose) mixes r_v (29%), a_v (27%), and r_d (24%). Together they capture 97% of the information.

near 3.0; adding proton data leaves the final value essentially unchanged (2.9), because protons largely reinforce the same dominant directions rather than opening new ones. The Igo ambiguity governs the global constraint structure.

Connection to Bayesian analysis.—The Fisher information matrix provides a rigorous foundation for the empirical findings of King *et al.* [16], who applied Bayesian MCMC to nucleon- ^{48}Ca , ^{90}Zr , and ^{208}Pb elastic scattering (9 free central parameters, fixing spin-orbit). They found that only V and r_v showed strong correlation, while all other parameter pairs appeared uncorrelated, in stark contrast to the frequentist χ^2 approach, which

produced spurious correlations among many parameter pairs.

In the Gaussian (Laplace) approximation [29], the posterior covariance equals $\Sigma_{\text{post}} = F^{-1}$: the FIM eigenvalues λ_i are the inverse posterior variances along principal axes. The eigenvector decomposition (Fig. 3) shows that $\mathbf{e}_1$ is dominated by r_v and V with the same sign (the volume integral direction), producing precisely the elongated V - r_v correlation King *et al.* observed. The remaining parameters contribute to sloppy directions with eigenvalues roughly 50 to 5×10^7 times smaller, rendering their correlations invisible in finite MCMC samples. The condition number of 10^6 or larger [Fig. 2(c)] also explains the spurious frequentist correlations: projecting an anisotropic “cigar-shaped” posterior onto a Gaussian tangent plane creates artificial coupling between the elongated and compressed directions.

The two approaches are complementary: Fisher maps the global constraint structure with $\sim 3700 \times$ fewer evaluations (see End Matter), while MCMC captures non-Gaussian features.

Discussion and Conclusion.—The results above answer the three questions posed in the Introduction. (1) Within the KD02 optical model framework, single-energy elastic scattering constrains only $D_{\text{eff}} \approx 1.7$ out of 13 parameter combinations; systematic multi-energy fitting raises D_{eff} to only ~ 2.3 – 2.5 on average (up to ~ 3 for the lightest systems) in the 13-parameter local space, and $D_{\text{eff}} \approx 2.9$ out of 48 in the universal coefficient space. (2) Within the local Fisher description, uniform error reduction rescales all eigenvalues without changing their participation ratio: D_{eff} measures the shape of the spectrum, not its total magnitude. Breaking the bottleneck therefore requires data that probe qualitatively different directions (systematic multi-energy fitting, spin

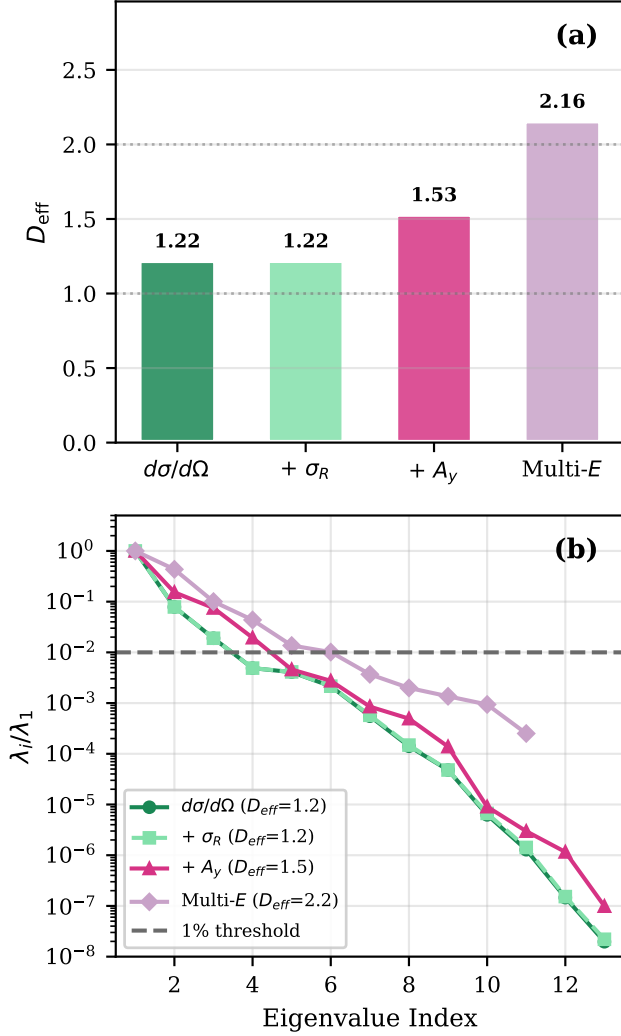


FIG. 4. Step-by-step constraint analysis for $n+^{40}\text{Ca}$. (a) D_{eff} for 13 local parameters increases as observables and energies are added (Steps 1–3 at 50 MeV; Step 4 combining 7 energies 10–200 MeV via Jacobian projection through the KD02 universal coefficient space): elastic alone (1.22), $+\sigma_R$ (no change), $+A_y$ (1.53), multi-energy with all observables (2.16). (b) Eigenvalue spectra at each step. The multi-energy curve has 11 non-zero eigenvalues because the KD02 constraints $r_w = r_v$ and $a_w = a_v$ eliminate 2 of the 13 local degrees of freedom.

observables), not simply smaller errors within the same error model. This is consistent with long-standing practice: physical interpretations of optical-model parameters are supplied by additional observables and by model structure, through matter densities folded with an effective nucleon-nucleon interaction [10] or dispersive relations linking real and imaginary parts across energies [6]. These inputs can constrain combinations that an isolated angular distribution leaves flat. A direct nucleon-nucleus example is the nonlocal dispersive analysis of ^{40}Ca : local and nonlocal absorptive potentials reproduce equiv-

alent elastic differential cross sections but give different angular-momentum absorption profiles, and only the nonlocal form also recovers particle number and charge density [30]. The resulting constraints on Woods-Saxon radii, diffusenesses, and depths remain conditional on the assumed density, folding interaction, dispersive ansatz, or global energy dependence, rather than being model-independent information contained in one angular distribution. The observable-class dependence found here has a well-known heavy-ion counterpart, where fusion and elastic scattering demand incompatible Woods-Saxon diffuseness values [31]. (3) Within the KD02 framework, the information bottleneck is stable: D_{eff} shows only weak dependence on target mass ($A = 12\text{--}208$), beam energy (10–200 MeV), and projectile type, and persists in the full 48-dimensional universal coefficient space ($D_{\text{eff}} = 2.9$ out of 48). This behavior traces to the surface-dominated scattering produced by the short-range nuclear force: the near-collinearity of the normalized sensitivity directions $\hat{S}_V$ and $\hat{S}_{r_v}$ at all angles (Fig. 6) is a consequence of the $J_V \propto V r_v^3$ structure shared by Woods-Saxon-type local potentials; verification with other global parameterizations is a natural extension.

The step-by-step decomposition provides actionable guidance for opening independent parameter directions: multi-energy fitting gives the largest increase from the elastic baseline ($\Delta D_{\text{eff}} = 1.24$ and 1.03 for $n+^{40}\text{Ca}$ and $n+^{120}\text{Sn}$, compared to 0.31 and 0.56 from A_y at the Method weighting, all computed from the unrounded single-energy D_{eff} values), while in the median A_y opens more independent structure than σ_R at a single energy (the quantitative change depends on ϵ/δ_{A_y} ; see End Matter). This is a ranking of spectral broadening, not of total Fisher information, which can increase even when D_{eff} decreases. For rare-isotope beam facilities [15, 32] where beam time is precious, the results favor distributing measurements across energies when the goal is to separate parameter directions rather than only to tighten an already-stiff combination.

In logarithmic KD02 coordinates, these results exhibit the standard local Fisher-spectrum signature of a “sloppy model.” The same Fisher analysis also applies to nucleon-nucleon potentials and chiral effective field theory, where the low-energy constants play the role of the optical potential parameters. Microscopic optical potentials from ab initio theory offer a path to bypass the parametric degeneracy entirely, by constraining the potential from nuclear forces rather than from scattering data alone. In practical terms, an elastic fit with a small χ^2 does not by itself determine the optical input needed for a transfer or breakup calculation; the unconstrained directions must either be pinned by external physics or propagated as uncertainty to the reaction observable. The D_{eff} values established here provide a benchmark within the stated coordinates and error models: any potential whose observables share a similar sensitivity geometry will face

a comparable information bottleneck when constrained by scattering data alone.

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Numerical method.—The Schrödinger equation for each partial wave is solved using the Numerov algorithm with matching to Coulomb functions at large radius. For spin-1/2 scattering, the solver iterates over all (l, j) partial wave channels, with the spin-orbit potential entering through the Thomas form with factor of 2 ($1 \cdot \sigma = 2l \cdot s$). From the (l, j) -resolved S -matrix, the direct and spin-flip scattering amplitudes $f(\theta)$ and $g(\theta)$ are constructed, yielding $d\sigma/d\Omega = |f|^2 + |g|^2$, $A_y = 2\text{Im}(fg^*)/(|f|^2 + |g|^2)$, σ_R , and σ_T . Parameter sensitivities are computed using two-point central finite differences with step size $\delta = 0.01|p_i|$, verified to be fully converged: scanning δ over three orders of magnitude ($10^{-5}|p_i|$ to $10^{-2}|p_i|$) changes D_{eff} by less than 0.2%, and a five-point stencil yields identical D_{eff} to three to four significant figures. All results use 35 equally spaced angles from 5° to 175° ; increasing to 50 or 90 angular points changes D_{eff} by less than 0.002. The solver has been validated against FRESKO [33], achieving S -matrix agreement to $|\Delta S| < 0.002$ and cross-section agreement to $< 0.2\%$.

Robustness checks.—Four checks confirm that the conclusions are robust to analysis choices. (1) *Evaluation point:* Perturbing all 13 KD02 parameters simultaneously by $\pm 10\%$ around the reference values (10 random trials per system) yields D_{eff} in the range 1.1–2.5 for elastic scattering, with means within 0.25 of the reference values. The conclusion $D_{\text{eff}} \ll N_p$ is robust to the choice of evaluation point. (2) *Geometry constraints:* Enforcing the KD02 constraints $r_w = r_v$ and $a_w = a_v$ (reducing from 13 to 11 free parameters) shifts the elastic D_{eff} by -0.4 to $+0.5$ across the 168 configurations (an increase in 115 of them; $+0.14$ to $+0.31$ for the representative 50 MeV systems), leaving the conclusion $D_{\text{eff}} \ll N_p$ unchanged in either parameterization. (3) *Observable weighting:* The combined D_{eff} depends on ϵ/δ_{A_y} . A census over all 168 configurations shows that the A_y gain is precision-dependent and neutron-dominated: at $\epsilon = 5\%$ ($\epsilon/\delta_{A_y} = 1.7$), adding A_y raises D_{eff} in 113 of 168 configurations (neutrons 68/84, median gain $+0.16$; protons 45/84, median ≈ 0), while at $\epsilon = 10\%$ the count drops to 92 of 168. Where A_y lowers D_{eff} , the added information makes the eigenvalue spectrum more peaked rather than opening balanced new directions (cf. the global analysis, Fig. 5). Adding σ_R raises D_{eff} in all 168 configurations but with median gain 6×10^{-4} ; the only gains above 0.05 occur for sub-barrier proton scattering ($p+^{197}\text{Au}$ and $p+^{208}\text{Pb}$ at 10 MeV), where the elastic angular distribution is Rutherford-dominated and constrains the absorptive parameters only weakly, so that σ_R opens an absorption-sensitive direction. At representative precision ratios the median gains order as multi-energy $\gg A_y \gg \sigma_R$. (4) *Back-projection convention:* The local-space multi-energy values of Step 4 use the stated map $F_{\text{ref}} = J(E_{\text{ref}}) F_{\text{global}} J(E_{\text{ref}})^T$. Because J is 13×48 with rank

11 (the KD02 constraints $r_w = r_v$, $a_w = a_v$), no unique inverse transformation exists and the back-projected D_{eff} is convention-dependent: propagating the pseudo-inverse covariance instead ($\Sigma_{\text{loc}} = J F_{\text{global}}^+ J^T$, $F_{\text{eff}} = \Sigma_{\text{loc}}^+$) gives larger values on average (2.9 instead of 2.3 over the 24 projectile-nucleus combinations, larger in 23 of the 24, insensitive to the pseudo-inverse threshold over four orders of magnitude). Under both conventions the multi-energy combination remains the largest broadening of the constrained subspace (from a single-energy all-observable average of 1.6), and the 48-coefficient global result $D_{\text{eff}} = 2.9$ involves no back-projection.

Condition number and parameter extraction.—The condition number $\kappa = \lambda_{\text{max}}/\lambda_{\text{min}}$ of the Fisher information matrix has a direct local interpretation for optical potential fitting. Consider $n+^{40}\text{Ca}$ at 50 MeV, where $\kappa \approx 5 \times 10^7$. In the Gaussian approximation, the FIM eigenvalues are the inverse posterior variances along the principal axes, $\delta\theta_i \propto 1/\sqrt{\lambda_i}$, so the local precision ratio between the sloppiest and stiffest directions is $\sqrt{\kappa} \approx 7000$. This ratio is a diagnostic of the local anisotropy, not a claim that the quadratic approximation remains valid over a literal 7000% displacement; establishing the global range requires a profile-likelihood or model-manifold calculation. Uniformly reducing the assumed errors rescales the local ellipsoid but leaves this axis ratio and D_{eff} unchanged.

This hierarchy also explains a familiar frustration in optical model fitting: χ^2 minimizers converge rapidly along the stiff direction but wander erratically along sloppy directions, producing run-to-run variations in parameters like a_v and W even when the fitted cross sections are virtually identical. The FIM eigenvectors identify these ill-determined combinations explicitly, and could be used to reparameterize the fit in terms of stiff and sloppy coordinates, eliminating the numerical instability at its source.

KD02 parameter hierarchy.—The KD02 potential involves parameters at two distinct levels.

Level 1: Local parameters (per system).—At a specific (A, Z, E) configuration, the potential is fully determined by 13 local parameters: real volume (V, r_v, a_v), imaginary volume (W, r_w, a_w), imaginary surface (W_d, r_d, a_d), and spin-orbit ($V_{so}, W_{so}, r_{so}, a_{so}$). Of these, $r_w = r_v$ and $a_w = a_v$ in KD02, but I retain them as independent in the Fisher analysis to test whether data could distinguish them; enforcing these constraints (reducing to 11 parameters) shifts D_{eff} by -0.4 to $+0.5$ across the scan ($+0.14$ to $+0.31$ for the representative 50 MeV systems) without affecting the conclusion $D_{\text{eff}} \ll N_p$. The single-energy Fisher matrix is 13×13 .

Level 2: Universal coefficients (global model).—The KD02 energy and mass dependence parametrizes each local parameter as a function of A, Z , and E through an-

analytical formulas with fitted numerical coefficients. For example, $E_f^n = -11.2814 + 0.02646 A$ involves *two* independent coefficients ($E_{f,0}^n$ and $E_{f,1}^n$), and $r_v = 1.3039 - 0.4054 A^{-1/3}$ involves two more. Counting all such coefficients from Tables 10–11 and Eqs. (25)–(26), (31)–(33), (36)–(37) of Ref. [27] yields 48 universal parameters. Table I provides a complete accounting.

TABLE I. Complete inventory of KD02 universal coefficients, counted from Ref. [27]. Each coefficient is an independently fitted number in the global model. “Shared” parameters are identical for neutrons and protons.

Group	N	Parameters
Shared geometry	9	$r_{v0}, r_{v1}; a_{v0}, a_{v1}; r_{d0}, r_{d1};$ $r_{so,0}, r_{so,1}; a_{so}$
Shared depth	16	$v_{10}, v_{1,\text{iso}}, v_{1A}, v_4; d_{10}, d_{1,\text{iso}};$ $w_{20}, w_{21}; d_{20}, d_{2,\text{amp}}, d_3;$ $v_{so,10}, v_{so,11}, v_{so,2}; w_{so,1}, w_{so,2}$
Neutron-specific	10	$E_{f,0}^n, E_{f,1}^n; v_{2,0}^n, v_{2,1}^n; v_{3,0}^n, v_{3,1}^n;$ $w_{1,0}^n, w_{1,1}^n; a_{d,0}^n, a_{d,1}^n$
Proton-specific	13	$E_{f,0}^p, E_{f,1}^p; v_{2,0}^p, v_{2,1}^p; v_{3,0}^p, v_{3,1}^p;$ $w_{1,0}^p, w_{1,1}^p; a_{d,0}^p, a_{d,1}^p; r_{c,0}, r_{c,1}, r_{c,2}$
Total	48	

The distinction between levels matters physically. The full dataset of 168 configurations (12 nuclei $\times$ 7 energies $\times$ 2 projectiles) probes the 48 universal coefficients, yielding $D_{\text{eff}} \approx 2.9$ (out of 48). Combining different nuclei adds constraining power through the A -dependence of geometry parameters, but the Igo ambiguity still dominates: the dominant eigenvector of $F_{\text{universal}}$ is 70% r_{v0} and 24% $r_{so,0}$, both radius coefficients.

Relation to Bayesian analysis.—Here I provide additional technical details for the Fisher-Bayesian comparison discussed in the main text. King *et al.* [16] found that:

- Only the depth V and radius r of the real potential showed strong correlation in MCMC scatter plots;
- All other parameter pairs exhibited approximately circular (uncorrelated) scatter plots;
- The frequentist χ^2 approach produced spurious strong correlations among many parameter pairs;
- The frequentist 95% confidence intervals were unrealistically narrow (44–92% empirical coverage versus the nominal 95%).

The narrow frequentist confidence intervals (Table I of Ref. [16]) are predicted by the Fisher analysis. The frequentist approach samples from the Gaussian approximation $\theta \sim \mathcal{N}(\theta, F^{-1})$, valid only near the minimum. But with $D_{\text{eff}} \approx 1.7$, the χ^2 surface has ~ 11 nearly flat directions where the Gaussian approximation breaks down,

causing the frequentist method to underestimate the true parameter uncertainty along sloppy directions.

Computational advantages.—The Fisher approach offers several practical advantages over Bayesian MCMC for systematic studies:

1. *Efficiency:* Computing D_{eff} for one system requires $2N_p + 1 = 27$ evaluations of the scattering solver (for $N_p = 13$), taking a few seconds per configuration. King *et al.* required $\sim 100,000$ MCMC samples per system for converged parameter correlations. The ~ 3700 -fold speedup enables the systematic scan across 168 configurations, an analysis that would require $\sim 10^7$ total solver calls with MCMC.
2. *Analytical results:* The scale-invariance of D_{eff} (invariance under $F \rightarrow \alpha F$) is an exact mathematical property, whereas establishing this from MCMC would require running chains at multiple noise levels.
3. *Decomposability:* The Fisher matrix can be additively decomposed by observable ($F = F_{\text{elastic}} + F_{A_y} + F_{\sigma_R} + \dots$) at each energy, and combined across energies via the Jacobian projection $F_{\text{global}} = \sum_E J(E)^T F(E) J(E)$ that accounts for the KD02 energy dependence (Step 4), enabling the step-by-step analysis in Fig. 4. Bayesian posteriors cannot be similarly decomposed without re-running the full MCMC.
4. *Subgroup analysis:* Block sub-matrices of F quantify the information content for parameter subgroups (real volume, imaginary, spin-orbit), providing physical insight into which observables constrain which parameters.

A natural next step is to use the FIM eigenvectors to define an optimal reparameterization that separates stiff from sloppy directions, potentially improving MCMC convergence by orders of magnitude [18].

Multi-system global analysis.—The per-nucleus KD02 analysis uses 13 local parameters specific to a given (A, Z, E) configuration. A natural question is whether combining data from many nuclei and energies can break the information bottleneck. To address this, I extend the analysis to the full KD02 universal coefficient space: 48 parameters (Table I) that generate all local potentials for any nucleus and energy through analytical prescriptions. These include shared geometry coefficients (e.g., $r_v = r_{v0} - r_{v1}A^{-1/3}$), energy-dependent depth amplitudes with Fermi-energy scales (E_f^n, E_f^p), and separate neutron/proton depth coefficients. For each of the 168 configurations (12 nuclei $\times$ 7 energies $\times$ 2 projectiles), the log-space Jacobian $J_i = \partial \log \theta_{\text{local}} / \partial \log \theta_{\text{universal}}$ transforms the local Fisher matrix into the universal

space, and the combined Fisher matrix is obtained additively:

$$F_{\text{universal}} = \sum_{i=1}^{168} J_i^T F_i^{\text{local}} J_i. \quad (4)$$

The central result of this analysis (Fig. 5) is that the information limit persists at the global level: $D_{\text{eff}} = 2.9$ out of 48 universal parameters. Figure 5(a) shows how D_{eff} evolves as systems are added cumulatively. With neutron data alone, the effective dimensionality rises sharply with the first few systems and peaks at $D_{\text{eff}} \approx 3.1$, at which point three independent parameter directions are being constrained, before settling near 3.0. Adding proton systems leaves the final value essentially unchanged ($D_{\text{eff}} = 2.9$), because the proton data largely reinforce the same dominant directions (primarily r_{v0}) that the neutron data already constrain, making the eigenvalue spectrum more peaked rather than broadening it. The final $D_{\text{eff}} = 2.9$ confirms that the information bottleneck is structural: the entire nuclear chart of elastic scattering data constrains only ~ 3 effective parameter combinations out of 48.

The eigenvalue spectrum [Fig. 5(b)] reveals an extreme hierarchy: the first eigenvalue $\lambda_1 = 4.2 \times 10^8$ carries 44% of the total Fisher information, the second (λ_2 , 37%) and third (λ_3 , 8%) add progressively less, and the top 4 modes capture over 90% of all information, while 9 modes reach 99%. The remaining 39 eigenvalues share only $\sim 1\%$ of the total, spanning 9 orders of magnitude below λ_1 . This near-singular structure means that the 48-dimensional parameter space is effectively collapsed to a ~ 3 -dimensional subspace by the data.

The eigenvector composition [Fig. 5(c)] reveals the physical identity of these constrained directions. The percentages below refer to squared components $|e_{i,k}|^2$, which give the fractional contribution of each parameter to a given mode (the figure displays the signed components, normalized so that the largest absolute value in each row equals 1). The dominant eigenvector $\mathbf{e}_1$ mixes r_{v0} (70%) and $r_{so,0}$ (24%), the leading coefficients in the real-volume and spin-orbit radius parametrizations. The second eigenvector $\mathbf{e}_2$ reverses these weights ($r_{so,0}$ 69%, r_{v0} 25%), while $\mathbf{e}_3$ is 91% r_{d0} (the surface imaginary radius). This confirms that the Igo ambiguity, which governs single-system fits, persists in the global constraint structure: elastic scattering across the nuclear chart primarily constrains the nuclear radius scale. All three leading modes are radius-dominated: scattering data constrain radii far more effectively than depths or diffusenesses.

Elastic-only data yield slightly *higher* D_{eff} than the all-observable result. This counterintuitive result illustrates a key distinction: D_{eff} measures the *shape* of the eigenvalue spectrum (how uniformly information is distributed), not the total information content ($\text{Tr } F$, which

always increases with more data). Adding A_y and σ_R increases the leading eigenvalues by factors of roughly 250–500, but deposits this information along already-stiff directions (predominantly r_{v0}) rather than illuminating new sloppy directions. The eigenvalue distribution becomes more peaked and D_{eff} decreases. The route to breaking the information bottleneck is therefore not simply “more data of the same kind,” but data that probe different aspects of the potential. Among the strategies tested in the main text (Fig. 4), only systematic multi-energy fitting effectively opens new constraining directions, because different beam energies sample different radial regions of the potential.

Angle-resolved sensitivity and information geometry.—

To understand *why* the information bottleneck arises, I examine the angular structure of parameter sensitivities. Figure 6(a) shows the normalized sensitivity directions $S_i(\theta)/\|\mathbf{S}_i\|$ for three parameters of the KD02 potential, computed for $n+^{40}\text{Ca}$ at 50 MeV. Normalizing each sensitivity vector to unit length removes the overall magnitude and isolates the angular *shape*: overlapping curves indicate collinear (degenerate) parameters, while separated curves indicate independent information content. Since D_{eff} is scale-invariant (ϵ^{-2} cancels in the participation ratio), this representation directly visualizes what determines D_{eff} .

The normalized sensitivities $\hat{S}_V$ (green) and $\hat{S}_{r_v}$ (pink) overlap almost perfectly across all angles, a direct visualization of the Igo degeneracy [8]: fractional changes in V and r_v produce angular distributions that are nearly indistinguishable, making these two parameters impossible to separate from elastic scattering alone.

The diffuseness $\hat{S}_{a_v}$ (lilac) overlaps with V and r_v at forward angles ($\theta < 100^\circ$). At backward angles ($\theta \geq 100^\circ$), a_v diverges and even changes sign, showing that diffuseness does develop an independent angular structure there. Because the Fisher matrix is built from logarithmic derivatives, the absolute magnitude of the cross section cancels in every matrix element, and backward-angle points are not down-weighted by their smaller cross sections; the root-mean-square logarithmic sensitivity of r_v in fact grows from 5.7 at $\theta < 100^\circ$ to 16.5 at $\theta \geq 100^\circ$. The reason the independent structure of a_v contributes little to D_{eff} is collinearity, not magnitude: 96% of the squared norm of the backward-angle a_v sensitivity lies in the plane spanned by $\mathbf{S}_V$ and $\mathbf{S}_{r_v}$, and the orthogonal remainder carries Fisher information of only 2×10^{-3} of the leading eigenvalue λ_1 . Restricting the Fisher matrix to backward angles alone gives $D_{\text{eff}} = 1.20$, essentially identical to the forward-angle value of 1.19 and the full-coverage value of 1.22: the degeneracy is present in every angular region separately, rather than being an artifact of how the regions are weighted. The constraint on diffuseness comes primarily from multi-energy systematics (Step 4), not from backward-angle coverage at a single energy.

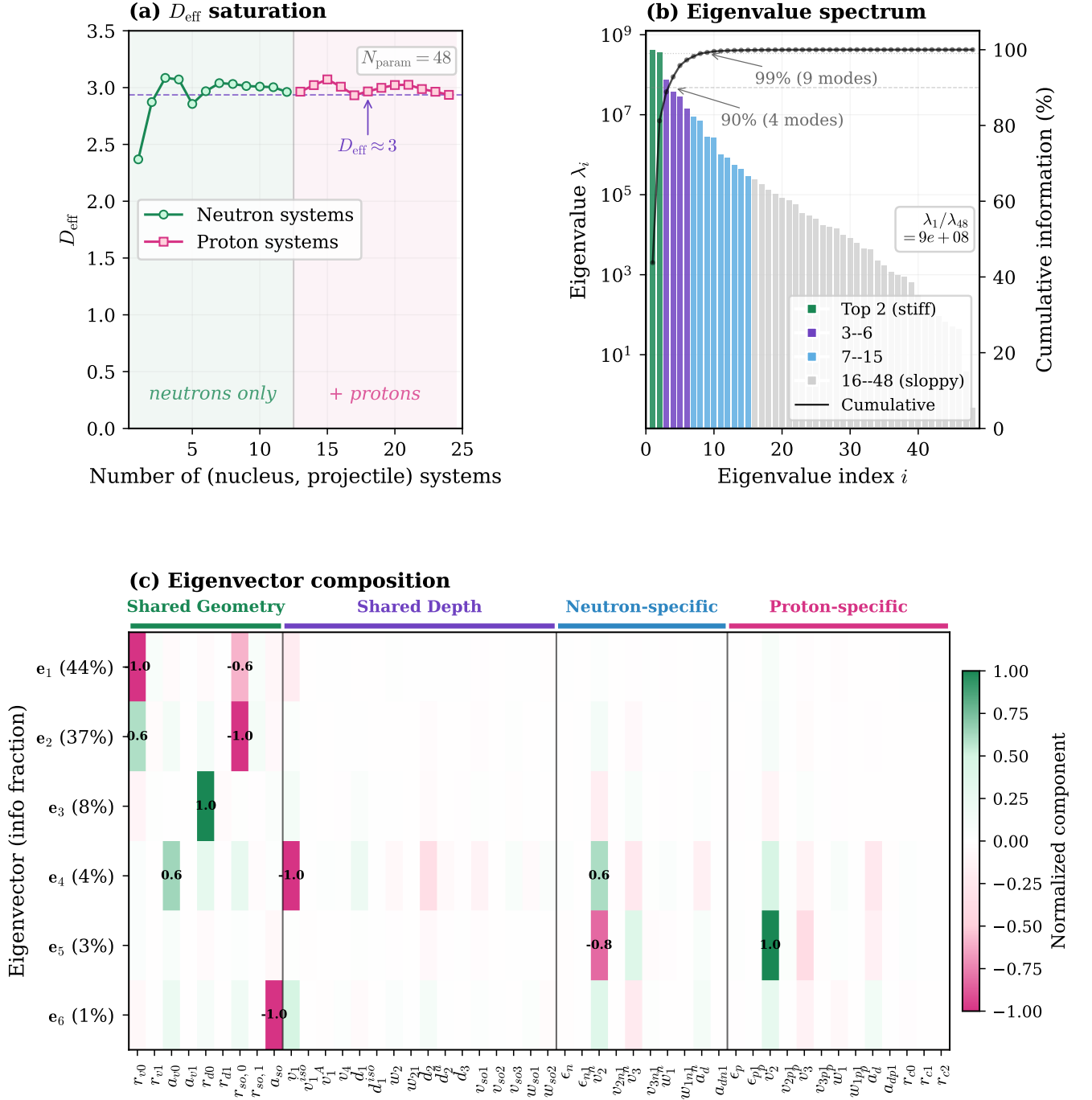


FIG. 5. Global Fisher analysis in the 48-dimensional universal KD02 coefficient space. (a) D_{eff} evolution as systems are added: neutron data (green) peak at ~ 3.1 after the first few systems and settle near 3.0; adding proton data (pink) leaves the final value essentially unchanged (~ 2.9), because protons largely reinforce existing stiff directions rather than opening new ones. (b) Eigenvalue spectrum: 4 modes carry over 90% of all information, 9 modes suffice for 99%, with condition number $\lambda_1/\lambda_{48} \approx 9 \times 10^8$. (c) Eigenvector composition of the top 6 modes, with components normalized so that the largest absolute value in each row equals 1. The y-axis labels give the information fraction carried by each mode. See text for the squared-component decomposition ($|e_{i,k}|^2$) of each eigenvector.

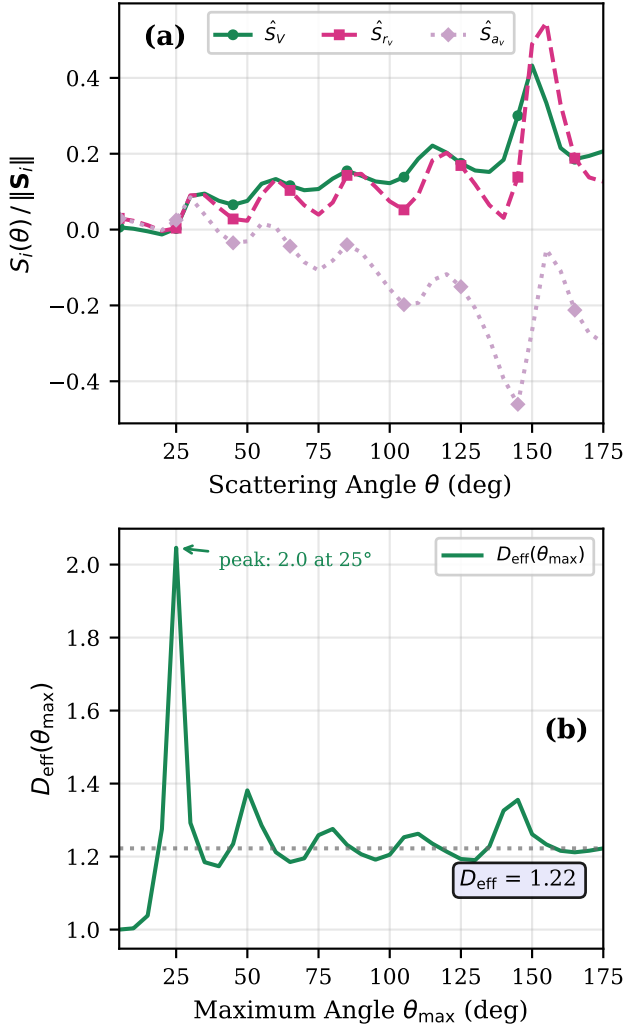


FIG. 6. Angle-resolved sensitivity for $n+^{40}\text{Ca}$ at 50 MeV. (a) Normalized sensitivity directions $S_i(\theta)/\|S_i\|$: overlapping curves indicate collinear (degenerate) parameter directions, while separated curves indicate independent information. V (green) and r_v (pink) overlap almost perfectly at all angles (Igo degeneracy). The diffuseness a_v (lilac) also overlaps at forward angles but diverges at backward angles ($\theta \geq 100^\circ$), showing that a_v develops an independent angular structure there. However, this independent component is small: 96% of the squared norm of the backward-angle a_v sensitivity lies in the $V-r_v$ plane (see text). (b) Cumulative $D_{\text{eff}}(\theta_{\text{max}})$: peaks at ~ 2.0 at 25° , on the approach to the first nuclear diffraction minimum (near 31°), where the steep angular variation briefly separates the sensitivity directions, then decreases as backward-angle data concentrates information along the dominant $V-r_v$ direction, settling to $D_{\text{eff}} = 1.22$ at full coverage.

Figure 6(b) shows the cumulative effective dimensionality $D_{\text{eff}}(\theta_{\text{max}})$, computed from the Fisher matrix integrated over $\theta \in [5^\circ, \theta_{\text{max}}]$. The curve is sharply non-monotonic: D_{eff} peaks at ~ 2.0 at $\theta_{\text{max}} = 25^\circ$, on the approach to the first nuclear diffraction minimum (located near 31°), where the steeply varying diffraction structure provides the most diverse constraints. As backward angles are included, information accumulates predominantly along the dominant $V-r_v$ direction, making the eigenvalue spectrum more peaked and reducing D_{eff} . The final value $D_{\text{eff}} = 1.22$ is reached at full angular coverage. This reveals an important subtlety: D_{eff} can temporarily *exceed* its final value at restricted angular ranges, because the forward-angle diffraction pattern distributes information more uniformly across parameters than full-coverage data. Extending angular coverage increases the *total* Fisher information ($\text{Tr } F$) but concentrates it more heavily along the dominant direction, thereby reducing D_{eff} .